

Meeting 1 – 16/02/2026

Attendees: Swapnil sir, Chilaka, Devadatta, Govardhan

Discussion Points:

- Discussed the overall **project proposal**.
- Discussed the work done on using Regex for identifying people's names as they are in standard format in all docs.
- Sir suggested focusing on **one specific task** due to limited time.
- Suggested task: **Identify ministries involved in a given text**, even when the ministry name is **not explicitly mentioned**.
- Avoid large tasks like:
 - Summarization
 - Sentiment analysis
 - Other complex NLP tasksbecause they require **annotated data and heavy evaluation**.
- Recommendation: **Choose a small task and study it deeply**.

Suggestions from Sir:

We should be able to explain:

- Why did we select this task?
- Why are we using a particular method?
- Why should the method work?
- **When and why the method might fail?**

TODO:

- Narrow the project scope to **ministry identification task**.
- Prepare a **revised proposal focusing on this task**.
- Think about **possible approaches and justification**.

Meeting 2 – 02/03/2026

Attendees: Swapnil sir, Chilaka, Devadatta

Discussion Points:

- Discussed the **updated proposal** focusing on **identifying ministries from text**.
- Explained the planned **methodology**:
 - Use **TF-IDF to identify ministry keywords**.
 - Map text to ministries using:
 - **Keyword matching**
 - **Vector embedding comparison model**.

Suggestions from Sir:

- Experiment with **different numbers of TF-IDF keywords**.
- Analyze:
 - How changing the number of keywords affects **probability distribution across ministries**.
 - Situations where the model **fails to identify the correct ministry**.
- Think about how to **use the prior knowledge** that only a few ministries (out of all 53) will be discussed through a session, **to modify/bias the probabilities?**

Evaluation Discussion:

- Use an **LLM (e.g., ChatGPT)** to label sample documents.
- Label **each paragraph with the ministry involved**.
- Compare **Model predictions vs LLM labels**
- Analyze differences:
 - Why they occur
 - Where the model performs well or poorly.

Documentation Requirement:

Clearly explain in the report, How the **reference data was collected**.

Sources used:

- **Parliament website PDFs** for parliament vocabulary.
- Information about **53 ministries** collected from:
 - Ministry websites
 - Wikipedia
 - LLMs.

TODO:

- Implement **TF-IDF keyword extraction for ministries**.
- Build **ministry knowledge base / vocabulary**.
- Test **different keyword counts**.
- Create a sample **dataset for evaluation**.
- Generate **LLM labels for comparison**.
- Document **data sources and methodology clearly in the report**.

Meeting 2 – 09/03/2026

Attendees: Swapnil sir, Chilaka, Devadatta

Discussion Points:

- Converted **53 ministries into vector embeddings** using **Sentence Transformer**.
- Each ministry is represented as a **384-dimension vector**.
- Visualized ministry vectors using 2D and 3D plots from **t-SNE** and **PCA**.
- The plots show **clear separation between many ministries**, indicating embeddings capture semantic differences.

TF-IDF Vocabulary Analysis:

- TF-IDF generated ministry vocabularies were analyzed using **frequency charts**.
- Observation:
 - Many words are **common across multiple ministries**.
 - These words belong to the **general parliament domain** and are not useful for distinguishing ministries.

Decision:

- Create a **domain-specific stopwords list** to remove common parliament words from TF-IDF vocabularies.

Suggestion from Sir:

- Use **Topic Modelling** to identify:
 - Topics in the documents
 - Important keywords within each topic.
- Concept discussed:
 - **Documents = distribution over topics**
 - **Topics = distribution over words**
- Suggested tool:
Online Topic Modeling Tool by David Mimno

Future Work:

- Later compare **classification performance using different approaches**, such as:
 - TF-IDF keyword based method
 - Embedding similarity method
 - Topic-based approach

TODO:

- Create **domain-specific stopwords list for parliament terms**.
- Clean TF-IDF vocabularies using this stopwords list.
- Experiment with **topic modelling** using David Mimno's tool.
- Extract **topics and keywords** from sample documents.
- Compare **results from different ministry classification approaches**.